WEATHERSAFE

Team Project Brief & Research Direction

A shared understanding of what we are trying to build, why it matters, what must be researched, and where the MVP boundary sits.

Working concept: a covert personal-safety application disguised as a normal weather app, allowing a user to silently initiate a safety incident and share selected information with trusted contacts.

Document purpose	Give all five team members one clear source of truth before research and design diverge.
Current stage	Concept definition, research, feasibility validation and early UX.
Primary output	A research-backed, technically credible prototype and vertical slice — not a production emergency service.
Method	Agile Scrum with iterative UX prototyping and early technical spikes.

1. Purpose of this brief

This document is the project lead's current definition of WeatherSafe. Its purpose is to reduce ambiguity before individual team members begin separate research. Everyone should be able to explain the same problem, target user, core journey, MVP, technical direction and project limitations. Research may change some decisions, but changes should be deliberate and recorded rather than introduced independently.

> **Rule for the team: research is not only about proving our idea is good. It must also identify where the idea is unsafe, unrealistic, unnecessary or technically infeasible.**

2. Project in one sentence

WeatherSafe is a covert safety app disguised as a weather app, enabling users to silently alert trusted contacts through concealed interactions while prioritising privacy through local processing and secure communication.

3. The problem we are addressing

Personal-safety technology usually assumes that a person can visibly unlock their phone, open an emergency feature, press a recognisable SOS control, make a call, or otherwise reveal that they are seeking help. That assumption can fail in situations involving harassment, stalking, intimidation, an unsafe date, unwanted following, coercion or another threatening interaction where openly asking for help may increase tension or expose the user's intentions.

Our project therefore focuses on a narrower question: can a safety application make requesting help less visible, while still giving the user meaningful control over what happens next and what information is shared?

The primary user group for our research is women who may experience harassment or threatening behaviour while travelling, socialising or going about daily life. This does not mean the finished concept must exclude other users; it gives the team a defined population around which to research the problem and design credible scenarios.

4. What we hope to accomplish

- Demonstrate a believable covert interaction model: the app should appear to be a normal weather application until the user intentionally activates a hidden safety function.
- Create a simple end-to-end safety workflow from covert trigger to trusted-contact awareness, rather than building disconnected features.
- Investigate whether on-device detection can support concealed gestures and/or codewords without making the system dependent on cloud AI.
- Explore a privacy-first communication model in which sensitive incident information is minimised, locally processed where possible, encrypted in transit and not routinely stored in a central cloud database.
- Produce a working vertical slice that proves the core concept and can be evaluated for usability, technical feasibility, privacy and reliability.

- Document our design decisions, research evidence, trade-offs and limitations so the final project can explain not only what was built, but why.

5. Intended user experience

The disguise is not a decorative theme. It is part of the safety mechanism. To an observer, WeatherSafe should behave like an ordinary weather application. The user should not need to navigate to an obvious page labelled Emergency, SOS or Safety in order to begin the core workflow.

Core journey

1. During setup, the user chooses trusted contacts and configures the permitted safety actions and information-sharing preferences.
2. In ordinary use, the application opens and functions as a believable weather app.
3. If the user becomes uncomfortable or unsafe, they use a concealed trigger such as a defined gesture or codeword rather than an obvious SOS control.
4. WeatherSafe begins an incident state and carries out only the actions authorised for that trigger or incident level.
5. A chosen trusted contact receives the alert and relevant information, such as the user's live or recent location and incident status.
6. The trusted contact can view a simple dashboard/timeline that helps them understand what is happening and respond appropriately.
7. The user can end, cancel or review the incident when it is safe to do so.

Possible graduated response model

A useful design direction is to avoid treating every uncomfortable situation as the same emergency. Research should test whether a graduated model is understandable and useful. For example: preparation/monitoring, check-in, trusted-contact alert, and higher-severity escalation. Exact levels and actions are not fixed yet and must be justified through research and feasibility testing.

6. MVP: what must exist for the project to succeed

MVP capability	Why it matters	Minimum proof
Weather-app disguise	The project's central safety premise is covert interaction.	A believable weather interface with no obvious safety controls on the normal surface.
Covert activation	The user must be able to start the workflow without visibly pressing SOS.	At least one reliable concealed trigger implemented and tested.
Trusted-contact alerting	The trigger is only useful if another person becomes aware.	A second device/contact receives an incident notification or session.
Location sharing	The contact needs actionable context.	Current/recent location is shared only when authorised by the incident flow.
Incident dashboard/timeline	The contact needs to understand status, not just receive a one-off message.	A simple view of incident start, status, location and key events.

Features that are valuable but not MVP dependencies

- Multiple ML-based gestures and advanced gesture classification.
- Automatic codeword recognition from continuous audio.
- Automatic evidence capture.
- Simulated/fake incoming call as an exit mechanism.
- Location-sensitive 'uncomfortable zones' that automatically alter monitoring behaviour.
- Automatic escalation to emergency services.
- A complex multi-level incident model.

These may become stretch goals or research findings. The team should not allow them to block delivery of the core vertical slice.

7. Technical direction — hypotheses to validate, not promises

The current architecture is intentionally ambitious. Research must distinguish what is desirable from what is practical on Android/iOS within the project timeframe.

- Cross-platform mobile application: Flutter or React Native are current candidates, but the choice must consider access to sensors, location, background execution, notifications and on-device ML.
- On-device ML: investigate lightweight gesture and/or keyword recognition using TensorFlow Lite/LiteRT or platform-native ML. Local inference is preferred where it improves privacy and responsiveness.
- Secure device-to-device communication: WebRTC is a candidate for peer-to-peer transport. The team must research signalling, NAT traversal, offline behaviour, reconnection and whether some minimal server infrastructure is unavoidable.
- Location: investigate permission models, foreground/background restrictions, update frequency, battery cost and user control.
- Security: authentication, trusted-contact pairing, encryption, key management, replay/spoofing risks and secure local storage must be considered from the start.
- Data minimisation: do not collect or retain sensitive data merely because it is technically possible.

> **Important: 'no cloud backend' and 'fully peer-to-peer' are research goals, not claims we should make until the technical lead proves the architecture is viable.**

8. Privacy, ethics and safety boundaries

Because this project handles location, possible audio, trusted contacts and safety incidents, privacy is a core functional concern rather than a final compliance checklist. The design should follow data minimisation, purpose limitation, user control, privacy by design and secure-by-default principles.

- The user should understand what information a trigger will share and with whom.
- Location sharing should be explicit within the configured safety workflow and should stop when the incident ends unless the user chooses otherwise.
- Sensitive data should be processed locally where practical and encrypted when transmitted.

- Any microphone or audio feature requires particularly careful consent, operating-system feasibility and legal/ethical review.
- The project must not imply guaranteed rescue, police dispatch or emergency-service integration unless such functionality is actually implemented and validated.
- False positives and false negatives matter. A safety feature that triggers unpredictably can itself create risk.
- Accessibility should include readable layouts, contrast, alternatives to audio-only activation, and consideration of users with motor or sensory impairments.

9. Research workstreams for the five-person team

Each workstream has a different question to answer. Team members should not simply collect links; each area must end in findings, decisions and implications for the product.

A. Project Lead / Product Integration (Ben)

Keep the team aligned on the product vision, research questions, MVP and backlog. Integrate findings rather than replacing the specialist work.

Research questions / tasks

- Maintain this brief and the decision log.
- Define/adjust MVP and acceptance criteria.
- Turn research findings into product decisions and backlog items.
- Prepare pitch narrative: why, who, what, how, differentiation and methodology.
- Resolve conflicts between desirable features, safety, feasibility and time.

Expected outputs

Updated product brief • MVP/backlog • Decision log • Pitch structure and project roadmap

B. Problem & User Research (Eyob)

Prove and define the problem before designing the solution. Focus on situations where visible help-seeking may be difficult or unsafe.

Research questions / tasks

- Find credible academic/official evidence on street harassment, stalking, intimidation, public-transport harassment, under-reporting and behaviour changes.
- Define the primary user group and important secondary users.
- Identify real contexts: walking, commuting, dates, nightlife, being followed, rides/taxis, etc.
- Identify current coping behaviours and barriers to seeking help.
- Convert evidence into 2–3 personas and clear user needs without inventing unsupported behaviour.

Expected outputs

Research summary • Source/evidence table • Personas • Problem statement • User needs and risks

C. Competitor & Reference Product Research (Priyanka)

Determine what existing safety products already solve, where our concept overlaps, and whether our proposed differentiation is real.

Research questions / tasks

- Investigate bSafe, Life360 and at least two additional relevant systems such as Noonlight or Red Panic Button.
- Verify activation methods, location sharing, evidence/recording, trusted contacts, subscriptions, cloud dependence, privacy model and emergency escalation.
- Identify what users must visibly do during an incident.
- Separate verified facts from marketing claims.
- Create a feature matrix and identify genuine gaps.

Expected outputs

Competitor profiles • Feature comparison matrix • Differentiation statement • Lessons to adopt/avoid

D. UX/UI & Safety Journey Research (Ben)

Design an interaction that remains understandable under stress while preserving the weather disguise.

Research questions / tasks

- Create scenarios for different severity levels.
- Map the complete user journey for both the person at risk and the trusted contact.
- Explore concealed trigger options and how users learn/remember them.
- Design cancellation, accidental-trigger and incident-ending flows.
- Create low-fidelity wireframes for weather view, setup, trusted contacts, incident state and contact dashboard.
- Evaluate whether the disguise creates discoverability or accessibility problems.

Expected outputs

User journeys • Use cases • Functional requirements • Wireframes • Prototype test plan

E. Technical Architecture & Feasibility Research (Ikram)

Objective: Determine whether WeatherSafe can realistically be developed using the proposed technologies, identify technical limitations early, and recommend a practical architecture for the MVP.

Research questions / tasks

- Compare Flutter and React Native, focusing on native device capabilities, performance, platform compatibility and development complexity.
- Investigate and prototype covert activation methods using device sensors, gestures and lightweight on-device machine learning.
- Research TensorFlow Lite/LiteRT and other suitable frameworks for recognising gestures or codewords locally.
- Investigate WebRTC and peer-to-peer communication, including signalling servers, NAT traversal, connectivity and reliability.
- Research Android and iOS restrictions concerning background execution, location tracking, microphone access and notification delivery.
- Investigate how real-time location sharing could work between the user and trusted contacts.
- Assess battery consumption, latency, device compatibility and offline behaviour.

- Recommend a realistic technology stack and identify features that may need to be simplified or deferred.

Expected outputs

Architecture diagram • Technical feasibility report • Technology comparison • Proof-of-concept prototypes • Recommended technology stack • Technical limitations and risks

F. Security, Privacy & Ethical Research (David)

Objective: Investigate how WeatherSafe can protect sensitive user information, prevent misuse and operate responsibly, particularly because the application handles personal safety incidents and potentially sensitive location and audio data.

Research questions / tasks

- Investigate security requirements for storing, transmitting and accessing sensitive user information.
- Research encryption methods for protecting communication between users and trusted contacts.
- Investigate authentication, trusted-contact verification, secure pairing and access control.
- Develop a threat model covering unauthorized access, stolen devices, spoofed alerts, intercepted communications, compromised contacts and data leakage.
- Research GDPR requirements relevant to personal data, location tracking, audio recording, consent and data retention.
- Investigate privacy-by-design principles, including data minimisation, local processing and user control.
- Assess the ethical implications of covert recording, microphone activation and location monitoring.
- Research potential misuse of the application, including stalking, unauthorised tracking or coercive monitoring.
- Investigate security and safety implications of false activations, missed alerts and unreliable communication.
- Recommend appropriate security controls and privacy requirements for the MVP.

Expected outputs

Security requirements specification • Threat model • Privacy and GDPR assessment • Data-flow diagram • Security risk assessment • Ethical considerations report • Recommended security controls

E & F were 1 role, but I split them into 2 to split up the work a little more, but the work is closely related so the 2 people doing this will cross reference their work to see if it's compatible.

How the two areas work together

Area	Main question	Primary responsibility

E. Technical Feasibility	Can we build it?	Architecture, frameworks, ML, communication, mobile capabilities and performance
F. Security & Privacy	Can we build it safely and responsibly?	Encryption, access control, threats, GDPR, ethics and misuse prevention

For example, the technical researcher might establish that WebRTC can transmit location updates between devices, while the security researcher investigates how those updates should be encrypted, authenticated and restricted to authorized contacts

10. Questions every research area should answer

- What evidence did we find?
- How trustworthy and current is that evidence?
- What does it mean for WeatherSafe specifically?
- Does it support, weaken or change one of our assumptions?
- What decision should the team make because of it?
- What new risk or unanswered question does it create?
- Is this needed for the MVP, a stretch goal, or future work?

11. Reference products and intended differentiation

Reference products are not included merely to say that competitors exist. They give us design patterns, technical expectations and a benchmark against which we can test whether WeatherSafe is actually distinct.

Reference	What to study	WeatherSafe hypothesis to test
bSafe	SOS/voice activation, location sharing, recording/evidence, trusted contacts and emergency workflow.	Can a safety workflow be made more covert and more locally processed?
Life360	Continuous/family location sharing, alerts, circles and the trade-off between convenience and persistent location visibility.	Can location be shared only when needed rather than making constant tracking the default?
Noonlight / another validated reference	How a safety app handles activation, monitoring, escalation and user reassurance.	Can trusted-contact support and covert activation serve lower or ambiguous threat levels without immediately behaving like an emergency dispatch service?

12. What makes WeatherSafe stand out — if research supports it

- Covert by design: the user should not have to reveal that they are using a safety application.
- Privacy by design: local processing and minimal central data collection should be architectural goals, not optional settings.

- Graduated response: the concept should support uncomfortable or escalating situations, not only an obvious emergency.
- Two-sided experience: design for both the person at risk and the trusted contact who must interpret and respond to the incident.
- AI/ML as an enabling technology, not the product itself: machine learning should make covert activation possible or more reliable, not be added simply to claim an AI feature.

Feature	WeatherSafe	bSafe	Life360	Noonlight
Emergency alert	✓	✓	✓	✓
Location sharing	✓	✓	✓	✓
Evidence recording	✓	✓	—	—
Covert interface	✓	Partial	✗	✗
Gesture/codeword activation	✓	Partial	✗	Partial
On-device ML	✓	—	—	—
Peer-to-peer architecture	✓	—	—	—
Weather-app disguise	✓	✗	✗	✗

13. Development methodology

The project should use Agile Scrum with iterative UX prototyping. We have uncertainty in the user interaction, ML reliability, peer-to-peer feasibility, privacy constraints and mobile-platform behaviour. Those uncertainties should be tested throughout development rather than discovered near the end.

Research → Requirements → UX Prototype → Technical Spike → Sprint → Test → Review → Refine → Repeat

Suggested delivery sequence

1. Understand: problem research, reference products, personas, initial requirements and feasibility questions.
2. Design: user journeys, covert interaction concepts, low-fidelity wireframes, architecture and threat model.
3. Core vertical slice: weather interface, one covert trigger, trusted-contact pairing/alerting and location sharing.
4. Incident experience: dashboard, status/timeline, cancellation/end flow and reliability improvements.
5. Advanced experimentation: additional ML triggers, codeword recognition, fake call or evidence features only if core flow is stable.
6. Evaluation: usability, security/privacy review, performance, false-trigger behaviour and final refinement.

14. Definition of the first vertical slice

The first convincing demonstration should be intentionally narrow: one user opens a normal-looking weather app, performs one concealed trigger, an incident starts, and one trusted contact receives the alert/location and can view the incident state. If the team can demonstrate that reliably, we have proved the core proposition. Everything else can then be layered on.

If time or feasibility forces cuts, preserve the end-to-end flow and remove sophistication. A simple reliable trigger connected to a simple reliable contact dashboard is more valuable than several impressive but disconnected ML prototypes.

15. Decisions that are NOT final yet

- Flutter versus React Native.
- Exact covert gesture(s) or codeword mechanism.
- Whether speech recognition belongs in the MVP.
- Whether true peer-to-peer communication can meet reliability requirements without any supporting backend/signalling service.
- Exact incident levels and escalation rules.
- Whether simulated calls, evidence recording or geofenced monitoring are appropriate and feasible.
- How much information is shown on the disguised weather interface during an active incident.
- Whether any emergency-service integration is within scope.

16. Out-of-scope / avoid promising

- A guaranteed emergency response service.
- Guaranteed police or ambulance dispatch.
- Continuous hidden surveillance of other people.
- Always-on microphone monitoring without explicit, defensible consent and legal/ethical justification.
- Perfect ML recognition or zero false activations.
- A claim that the system is fully P2P/no-cloud until the architecture is proven.
- A production-ready safety platform suitable for real-world dependence without extensive testing, security review and operational support.

17. Shared documentation standard

To prevent five separate research streams from becoming five separate versions of the project, all meaningful findings should feed a shared research-and-decisions log.

Date	Question	Evidence / research	Finding	Decision / implication	Source	Owner
07/10	Can P2P work reliably?	WebRTC research + spike	Signalling may still be required	Do not promise zero backend yet	Link/ref	Tech lead

The goal is traceability: later, when the team is asked why a feature, technology or design was chosen, the answer should be supported by research or prototype evidence rather than memory.

18. Early evidence already identified

The problem-research workstream can begin from two studies already identified in our earlier project discussion. These are starting points, not the complete literature review:

- Fleetwood, J. (2019), “Everyday self-defence: Hollaback narratives, habitus and resisting street harassment,” The British Journal of Sociology — analyses more than 500 first-person accounts of street harassment and women’s responses.
- A 2024 systematic review indexed on PubMed examining harassment against women in public transport across 28 studies — useful for evidence on frequency, under-reporting and changes to travel behaviour.

The research lead should retrieve the full bibliographic details, evaluate the studies, add complementary recent sources, and avoid stretching findings beyond what the papers actually support.

19. Current project success criteria

At this stage, the project should be considered successful if the team can:

- Clearly demonstrate that the chosen problem is real and that the proposed covert interaction addresses a meaningful gap.
- Explain who the primary user is and show realistic, research-informed scenarios.
- Deliver a believable weather disguise and a usable concealed activation method.
- Complete an end-to-end incident flow to a trusted contact.
- Show a defensible privacy/security architecture and openly state its limitations.
- Evaluate the prototype rather than merely demonstrate it.
- Show how research changed or validated the product decisions.
- Keep the final scope coherent enough that every feature supports the same central proposition.

> **North-star question: When openly asking for help may be unsafe, can WeatherSafe let a person ask for help silently, privately and reliably?**